

12. What is the maximum data rate supported by the FlexRay protocol?
 A) 5 Mbps B) 10 Mbps
 C) 20 Mbps D) 50 Mbps
13. What is the key feature of 1000BASE-T1 Automotive Ethernet?
 A) It operates over a single twisted pair cable at 1 Gbps B) It supports deterministic communication
 C) It uses optical fibers for high-speed communication D) It is only used for engine control units
14. Which physical layer technology is most commonly used in Automotive Ethernet?
 A) Single-Pair Ethernet (SPE) B) CAN transceivers
 C) Fiber optic cables D) Wireless transceivers
15. Automotive Ethernet is preferred for ADAS applications because it:
 A) Provides low-cost wiring solutions B) Offers high bandwidth and low latency
 C) Supports compatibility with LIN and CAN protocols D) Consumes less power than all other bus systems
16. What is a key advantage of Automotive Ethernet over traditional automotive networks like CAN or LIN?
 A) Lower implementation cost B) Ability to handle large data volumes and high-speed communication
 C) Exclusive use in safety-critical applications D) Compatibility with all existing vehicle systems
17. What is the maximum cable length typically supported by Automotive Ethernet at 100 Mbps?
 A) 15 meters B) 40 meters
 C) 100 meters D) 300 meters
18. Which type of messages benefit most from TSN in Automotive Ethernet?
 A) Low-priority messages B) Time-critical messages like ADAS data
 C) Diagnostic messages D) None of these
19. Which protocol allows Ethernet frames to carry diagnostic messages in Automotive Ethernet?
 A) DoIP (Diagnostics over IP) B) TCP/IP
 C) AVB D) UDP
20. What feature of Automotive Ethernet enables power and data transmission over the same cable?
 A) AVB B) PoE (Power over Ethernet)
 C) BroadR-Reach D) TSN

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) What are the benefits of AUTOSAR in the automobile sector?	4	L*2	1	1
	b) List the different goals of the AUTOSAR system.	4	L2	1	1
	c) Explain the important characteristics of the LIN bus system.	4	L2	1	1
	d) With a neat diagram, explain the physical layer in the SENT/SPC bus system.	4	L2	1	1
2.	a) With a neat diagram, explain the AUTOSAR layered architecture.	8	L2	1	1
	b) Explain the different functions of an Automotive Chassis.	4	L2	1	1
	c) What are the different applications of big-data in the automobile sector?	4	L2	1	1
Unit – II					
3.	a) Explain the important characteristics of the CAN bus system.	4	L2	2	1
	b) What are the key features of the CAN XL bus system?	4	L2	2	1
	c) With a neat diagram, explain the structure of the FlexRay frame format.	8	L2	3	1
4.	a) With a neat diagram, explain the network node architecture of the CAN bus system.	8	L2	2	1

b)	Explain the important characteristics of the FlexRay bus system.	4	L2	3	1
c)	With a neat diagram, explain the device model of the MOST bus system.	4	L2	3	1
5. a)	With a neat diagram, explain the basic CAN module system.	8	L2	2	1
b)	Outline the network node architecture of the FlexRay system.	4	L2	3	1
c)	With a neat diagram, explain the data frame of the MOST-25 bus system.	4	L2	3	1

Unit – III

6. a)	Explain the physical layer of BroadR-Reach ethernet used in the automotive sector.	8	L2	4	1
b)	Explain the different functionalities of the Time-Sensitive Networking (TSN) components.	8	L2	5	1
7. a)	Explain the passive and active ADAS applications of 100 Mbps BroadR-Reach ethernet.	4	L2	4	1
b)	Explain the physical layer of 1 Gbit/s Ethernet used in automobiles with a neat diagram.	4	L2	4	1
c)	Describe the three core components of the 802.1 Time-Sensitive Networking standards.	8	L2	5	1
8. a)	Explain the physical layer of 100BASE-T1 Ethernet used in automobiles with a neat and labelled diagram.	8	L2	4	1
b)	Outline the layers of the OSI model with Time-Sensitive Networking standards.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
